Township: Technical Design Document

Repository	https://github.com/AmanSingh-tech/township
Live app	https://township-91vm.onrender.com/welcome (Render free plan: the first visit after a quiet spell takes about a minute). Demo sign-ins are listed on the sign-in page.
Client	Company A, Pune: 2,400 flats in 28 towers
Author, date	Aman Kumar Singh, 4 November 2026

1. Context

Bills are made in Excel, payments are matched to flats by hand before Tally, complaints arrive by phone and 28 WhatsApp groups, and visitors go into paper registers. The brief puts the cost at ₹83.57 lakh a year. It targets on-time payments up from 70% to 90%, 80% less time spent matching payments, and reopened complaints down from 25% to under 8%.

Core build, shipped end to end in 8 screens, on the brief's sample data (its three flats are reproduced exactly and tested):

Screen	Role	Purpose
Sign in	All	Mobile number and a one-time code, linked to a flat
Resident home	Resident	Dues, open complaints and today's visitors; every action within two taps
Bills and test payment	Resident	Area × rate, late interest, test checkout, numbered receipt, ledger
Raise a complaint, complaint status	Resident	Text and photo; AI category and urgency; timeline; rating on closure
Help desk queue	Help desk	Auto-assignment, status updates, one-tap category change
Bank matching	Accounts	Bank CSV in; confident credits post, unclear ones are queued
Committee report	Committee	Collections, dues ageing, complaint times; signed and frozen monthly

Everything else that is built, as optional depth from the brief's "must have":

Who	Features
Residents	Installable web app on every screen size (an Install app button in every navigation bar: the browser's prompt on Chrome and Edge, the steps for Safari on iPhone, iPad and Mac; PNG and maskable icons, shortcuts, an offline app shell); family and tenant members; notices; guest codes with a countdown; amenity booking by slot with rules, fees on the bill and free cancellation until a cut-off; receipts as PDFs; push alerts; English, Marathi and Hindi; light and dark; consent, data download and erasure requests (DPDP)
Help desk	Technicians per category and tower; response timers; linked duplicates; notes to the resident; audited corrections to residents' details
Accounts	Monthly bill runs; register import from Excel with a preview; Tally export; two-person approval of manual posts, adjustments and login-number changes
Committee	Approves bill runs; the brief's targets on the report; sign-off that freezes a month with a SHA-256 fingerprint
Guards	Paired tablets with PIN sign-in; visitor photo, approval and guest codes; an offline queue; the DPDP notice on screen; visitor photos deleted after 90 days

Admin	Settings, staff, technicians, gate tablets, amenities, notices, the message outbox, data requests, the audit log
Everyone	A scroll-driven front page at `/welcome`; demo logins on the sign-in page; an integrity check (`cli verify`)

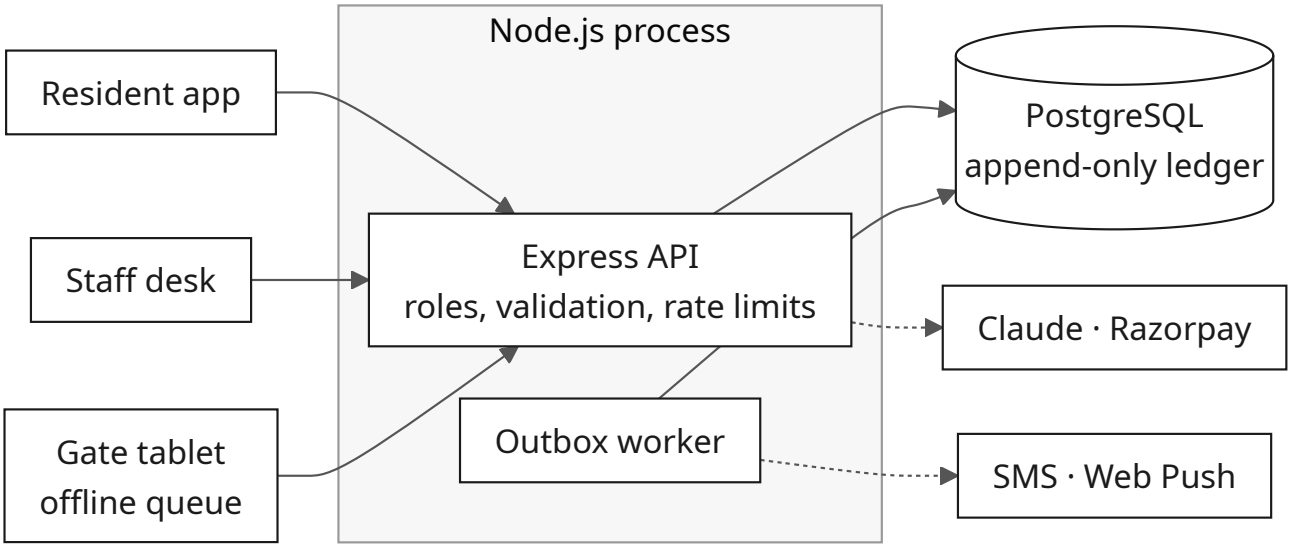
Out of scope, as in the brief: accounting beyond maintenance (Tally stays), CCTV and barriers, a marketplace.

2. Assumptions

Given by the brief: ₹6.50 per sq ft a month, late fees under the bye-laws, visitor photos deleted after 90 days, English, Marathi and Hindi, and mocked payments and messages. Where the brief is silent, I assumed the following. Each is a setting or a single function.

#	Assumption
A1	"Days overdue" counts from the oldest unpaid due date, so T12-0408 (47 days) also owes the next month: ₹12,788 plus ₹216 interest.
A2	Late fee means simple interest on overdue principal at 21% a year (the model bye-law ceiling). Payments clear interest first, then the oldest bill.
A3	Bills are raised on the 1st and due on the 10th. Opening arrears carry no interest in the app.
A4	Bank statements are CSV exports. UPI narrations carry the payer name, UPI ID and UTR, and sometimes a flat number.
A5	A credit posts automatically only on a flat number or registered UPI ID, a score of 80 or more and a lead of 30 over the next flat.
A6	Response targets are 15 min (critical), 2 h (high), 8 h (medium) and 24 h (low). The duty manager is one configured number.
A7	Roles beyond the core two: accounts, committee, admin and guard. Money decisions need two different people.
A8	Receipts are numbered without gaps per financial year, e.g. `RCT/2026-27/00001`.
A9	One server is enough: about 8 requests a second at the bill-day peak (most bills are paid in the first 10 days).

3. Architecture



React and Vite on the client; Node 20, Express and PostgreSQL 16 on the server; one Docker image. Dotted lines are paid services, mocked when no credentials are set.

Decisions

> **ADR-1 · One modular service, not microservices**
>
> **Context:** one township peaks at about 8 requests a second. **Decision:** a single Node process with clear modules (billing, matcher, complaints, notify). **Consequences:** one deploy and no network failures between services; the load test shows 270–300 requests a second on one process.

> **ADR-2 · PostgreSQL, with the ledger append-only in the database**
>
> **Context:** the committee needs a report it can trust. **Decision:** bills and payments write ledger lines in one transaction; triggers block updates and deletes, and the app's database role cannot disable them. **Consequences:** corrections are new adjustment lines, and an integrity check (`cli verify`) can prove the books balance.

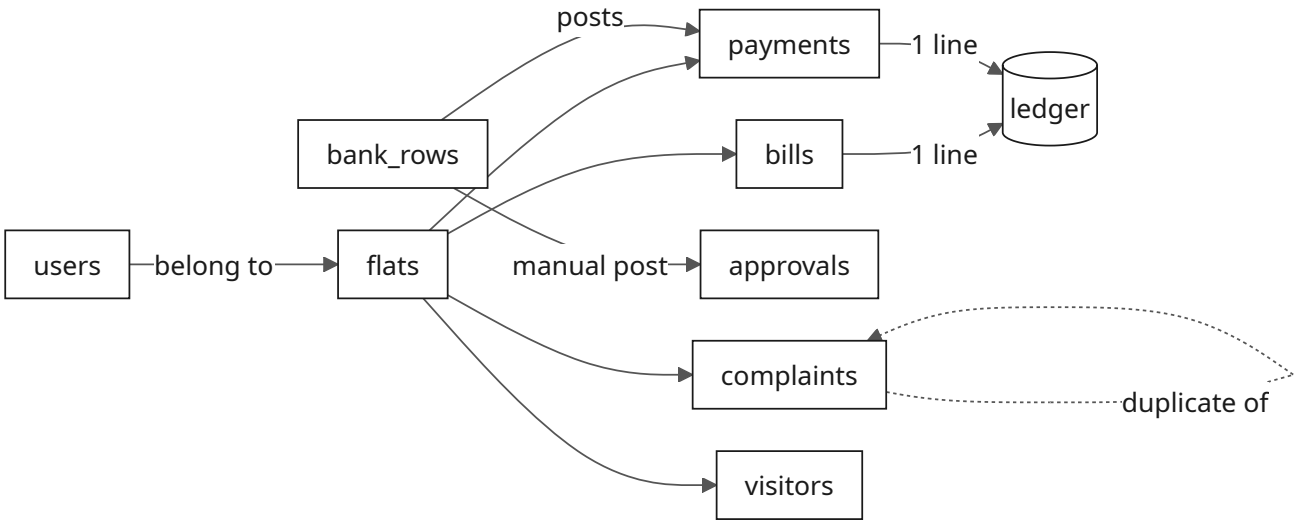
> **ADR-3 · Payment matching by explained rules, not a model**
>
> **Context:** a wrong match puts money on the wrong flat. **Decision:** score each credit with reasons; post only on strong, unambiguous evidence; queue everything else for a person, with a second person approving. **Consequences:** every post is repeatable and auditable; the share posted automatically grows as UPI IDs are remembered.

> **ADR-4 · Complaint triage by LLM, with a rule above it and a local fallback**
>
> **Context:** a wrong category is cheap to fix; a missed gas leak is not. **Decision:** safety words escalate by rule first; Claude classifies with a fixed schema; a local keyword model is used without a key or on any error. **Consequences:** a complaint is never blocked by the network, and the AI is never the last word on safety.

> **ADR-5 · Messages through a database outbox**
>
> **Context:** a crash must not lose a visitor alert or send one for a rolled-back payment. **Decision:** messages are written in the same transaction as their cause and sent by a worker, urgent first (`SKIP LOCKED`). **Consequences:** no extra queue to run; a visitor alert behind 4,200 bill alerts goes out in 0.9 s.

> **ADR-6 · An installable web app with one-time-code sign-in**
>
> **Context:** residents open the app for seconds and forget passwords. **Decision:** a web app installed from the browser, with push alerts; sign-in by a code to the registered number; guards use a PIN on a paired tablet. **Consequences:** no app store, and one codebase for phones and gate tablets.

4. Data model

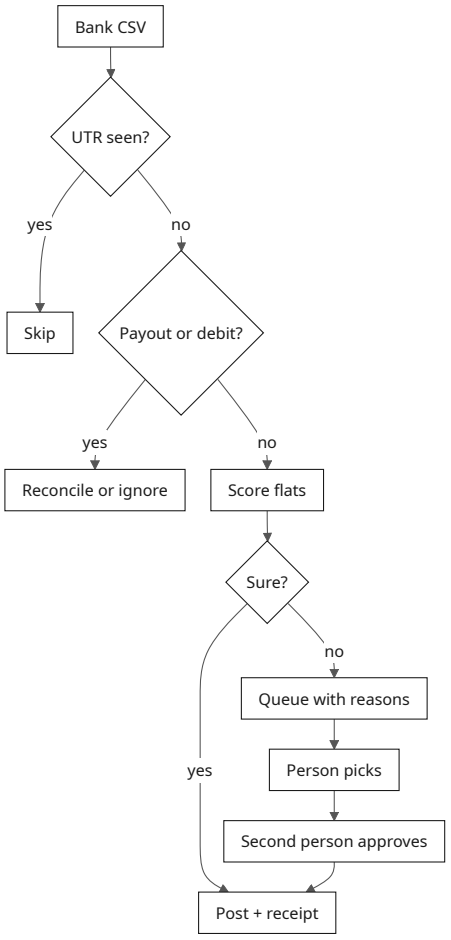


There are 34 tables in total; the core is shown. A flat's balance is always the sum of its ledger, never a stored number.

Rule	Enforced by
The books can't be rewritten	Triggers on `ledger`, `payments`, `audit_log`, `report_signoffs`, and a restricted app role
One bill per flat per month	`UNIQUE (flat_id, period)`, so a retried bill run can't double-bill
A gateway payment is recorded once	`UNIQUE gateway_payment_id`, so the callback and webhook are both safe
Nobody approves their own request	`CHECK (decided_by <> requested_by)`
Money is whole rupees	Integer columns with non-negative checks

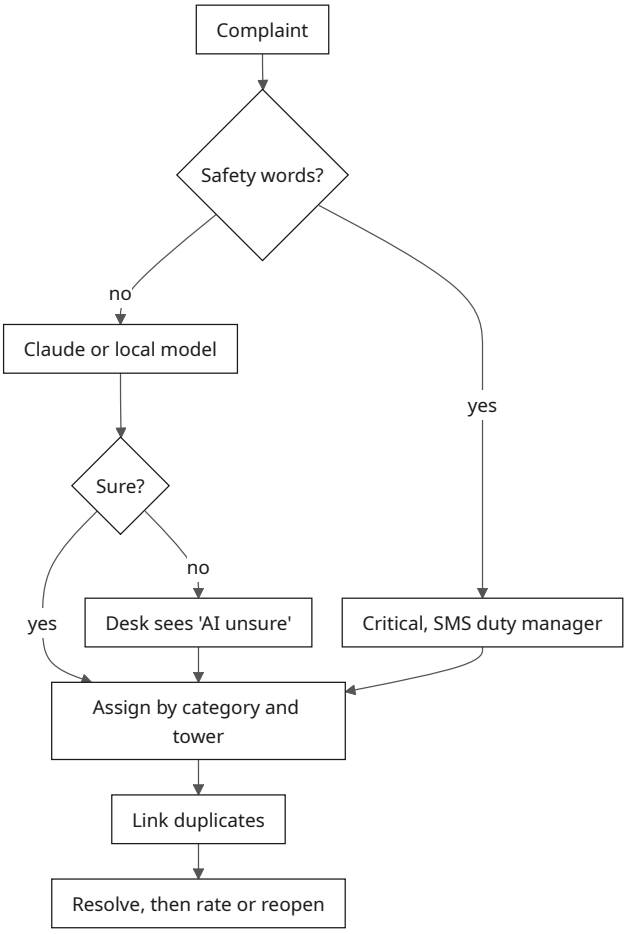
5. Main flows

Bank statement matching (the hard part)



"Sure" means a flat number or registered UPI ID, a score of 80 or more, a lead of 30 or more, and no other flat with strong evidence. Re-uploading a statement posts nothing new.

Complaint triage



"Sure" means the model's confidence is 0.6 or more. Safety words are fire, gas, lift stuck and shock, in three languages.

Payment: the server creates the gateway order for the amount it computes; after checkout it verifies the signature and fetches the payment from the gateway, then writes the payment, ledger line, receipt number and alert in one transaction. A signed webhook covers residents who close the app mid-payment. **Gate:** entries are saved on the tablet while offline and sent with client ids when the network returns, so nothing is duplicated.

6. AI and automation

The brief requires	Built as
Match bank rows by amount, reference and UPI ID	An evidence score: flat number in the reference +60, registered UPI ID +50, amount equals dues +30, payer name +15; far above dues −40
Unclear payments are never posted automatically	Name or amount alone never posts; conflicts are queued with the top three flats and reasons

Classify complaints by category and urgency	Claude with a fixed category list and JSON schema; refusals, invalid answers or an 8 s timeout fall back to the local model
Help desk changes a category in one tap	A category menu; below 0.6 confidence the complaint is flagged "AI unsure"; the AI's original guess is kept
Fire, gas, lift stuck or shock always escalate	A rule run before the AI, in English, Hindi and Marathi; urgency is set to critical and can't be lowered
Link duplicates, never close them	Reports of the same tower asset are linked to one job; each flat closes and rates its own

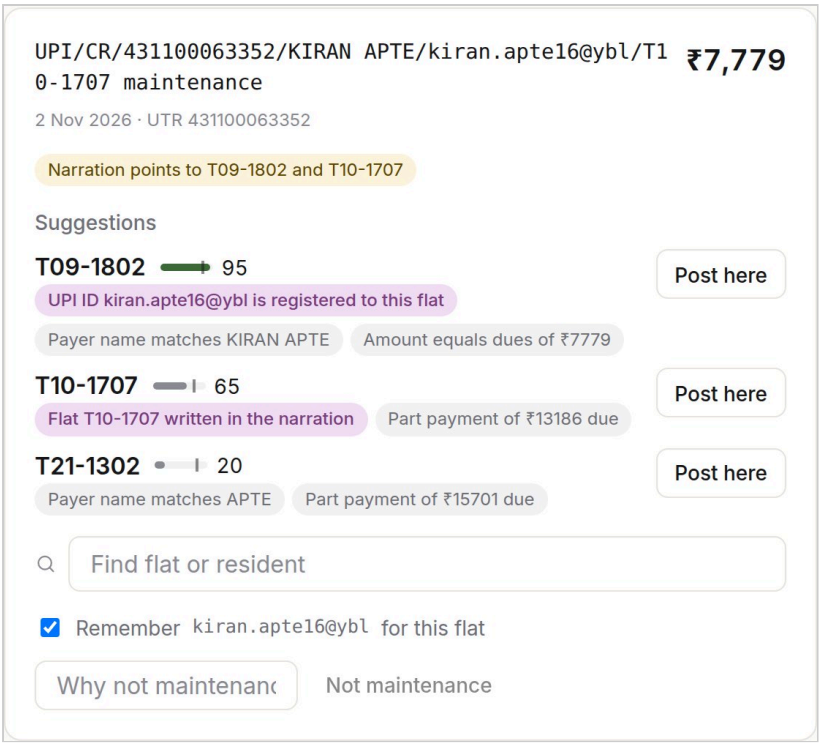
Measured, not assumed

`npm run eval` scores both automations and writes `docs/evaluation.md`; a test holds each promise below as a floor.

Complaint triage: 90 labelled complaints (EN, HI, MR, Hinglish)	**Local model**
Category right; right when it says it is sure	81%; 100% (35% of complaints)
Flagged "AI unsure" for the help desk to check	64%
Hazards escalated; false alarms	14 of 14; 0

Bank matching: 5 statements, 140 credits with known flats, 30 other rows	**Result**
Posted automatically; **posted to the wrong flat**	55%; **0**
Queued with the right flat as the top suggestion	98% (100% in the top three)
Matching time against doing it by hand (the brief's target: −80%)	−88% (assumed 2 min a row by hand, 30 s to confirm a suggestion)

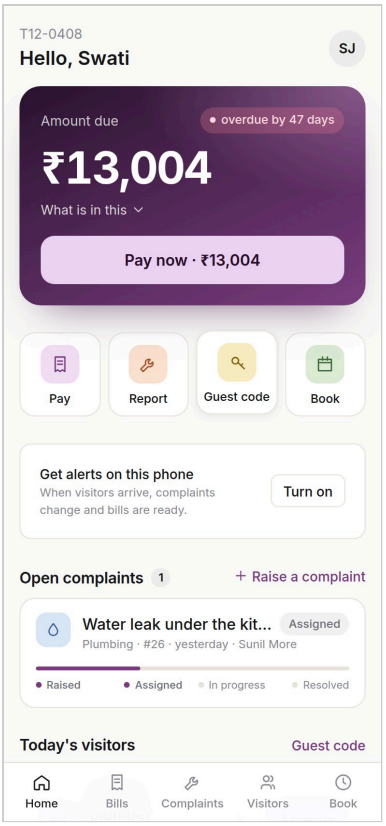
What the first run found. Hindi and Marathi words were being split on their vowel signs, so their keywords never matched (50% right; now 86% and 71%). The safety rule missed the Marathi "अडकले" (stuck) and raised alarms on "लिफ्ट बंद" (the lift is off) and "fire lane". All three are fixed. The local model is cautious: it flags most complaints as unsure, which costs the help desk a tap, never a wrong job. With a Claude key the same set runs with `--claude`.



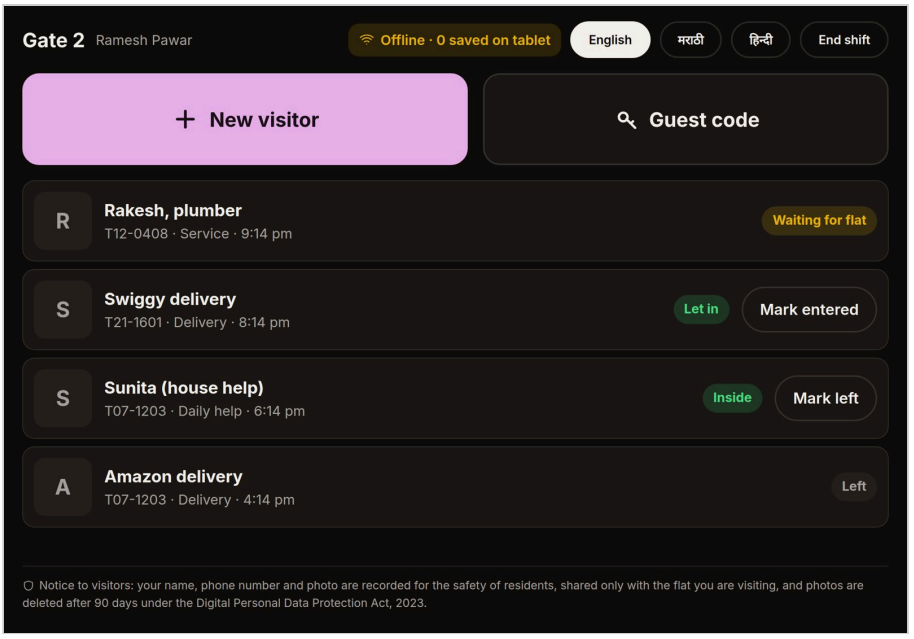
A credit whose narration names T10-1707 while its UPI ID is registered to T09-1802. Two strong signals disagree, so it is queued with both flats and their evidence instead of being posted.

7. UI and design

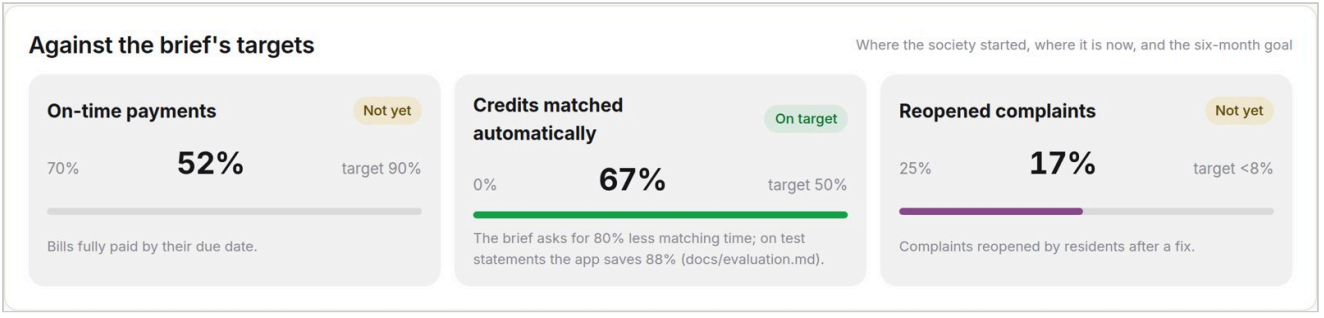
The brief asks for minimal design and every action within two taps. Warm neutrals with one lilac accent for the next action; six soft tints only to tell categories, people and amenities apart; light and dark themes; loading, empty and error states from shared components. Small interactions carry meaning: the dues card opens into what makes up the amount, a tapped month in Bills shows its sum, a waiting visitor pulses on the timeline, and guest codes count down. Animations stop when the phone asks for reduced motion. The logo is a T made from a tower between two shorter ones; the front page (`/welcome`) tells the product in seven chapters, each with its own colour, that arrive as sheets or through an opening circle; it is built with GSAP and loads on its own, so the app screens carry none of it. The desk is denser, for a working day, and the committee report leads with the brief's three targets.



Resident home: dues first, with the overdue days and interest stated; complaints and visitors one tap away.



Gate tablet with the network down: very large buttons, a dark screen for night, "Offline · saved on tablet", language switch, and the DPDP notice to visitors.



Committee report: the brief's success measures from where the society started, with the six-month target.

8. Security and testing

- **Sign-in:** codes stored as hashes, valid for 5 minutes with 5 tries, at most 3 per number an hour. Unknown numbers get the same reply as known ones.
- **Access:** every route declares its roles, and residents only ever reach their own flat. Inputs are validated on every route, and CSV exports can't run spreadsheet formulas.
- **Data:** no secrets in the repository (only `.env.example`), and photos are served through 15-minute signed links. Under the DPDP Act: consent, data download, erasure, and visitor photos deleted after 90 days.
- **Tests:** 53 server and 6 browser tests, all passing. They cover the brief's sample flats, interest, gapless receipts, every matching case, safety words in three languages, role access, two-person approvals, and the evaluation floors (no wrong posts, every hazard caught).
- **Load test** at full size (2,400 flats, a year of data): 95% of requests under 28 ms at five times peak; a bill run for every flat in 4.1 s; 2,400 bank credits matched in 17.5 s.

9. Cut, next and limits

- **Cut:** WhatsApp (needs Meta business approval), real SMS and payments (mocked, as the brief allows, following the real providers' steps), native apps, and technician sign-in.
- **Next:** grow the labelled set from the help desk's corrections (the AI's guess is stored with every complaint) and compare Claude on it; remember UPI IDs by default, the biggest lever on automatic matching (family members' IDs are 10% of credits); WhatsApp alerts through the same outbox; bank feeds instead of CSV uploads.
- **Limits:** Render free plan (sleeps when idle; uploaded photos are lost on restart); a keyword-based local classifier, with anything unsure going to a person; one server process.